Welcome! 👋

TOTALLY.

“What's a Meta-Law?” I hear you ask.

That's...ominous ?

"Our burn-out rate is only third in the Company, you know ?"

"You have to understand: you're part of the (corporate) Matrix, now..."

Conway's Law

> Any organization that designs a system will produce a system that mirrors how that organization communicates.

In other words…

🛑 Your architecture is already decided: just look at your teams

🤷 Every team has its habits and its own territory

🍪 “Culture eats strategy for breakfast.”

How can we work around Conway's Law?

👷 Build teams that work across the whole stack (DevOps, Full Stack, etc.)

🏴‍☠️ Let people switch teams to share good practices

🦄 Value a mix of skills (cross-functional task forces)

DEVOXX MOROCCO

Anyway… back to Laurie

“What if I deleted this bit of code? It looks like a real mess…”

"What have you doooooooone !"

Poor Laurie has just run into…

Chesterton's Fence

> Before changing something, first understand why it is the way it is.

In other words…

💰 “Legacy” projects pay the bills. Let's treat them with respect!

👵 “Legacy” starts at the first commit. Maintaining it is a good thing

😅 That “messy” code has a long history behind it. Understand it !

How can we deal with Chesterton's Fence?

🤔 **Good**: Take time to *understand* the domain (with DDD, for example)

📝 **Better**: Document the business rules yourself (automated tests! LLMs !)

💪 **Best**: Improve the “Legacy” code over time. We don't have excuses anymore !

Let's check in on Laurie again…

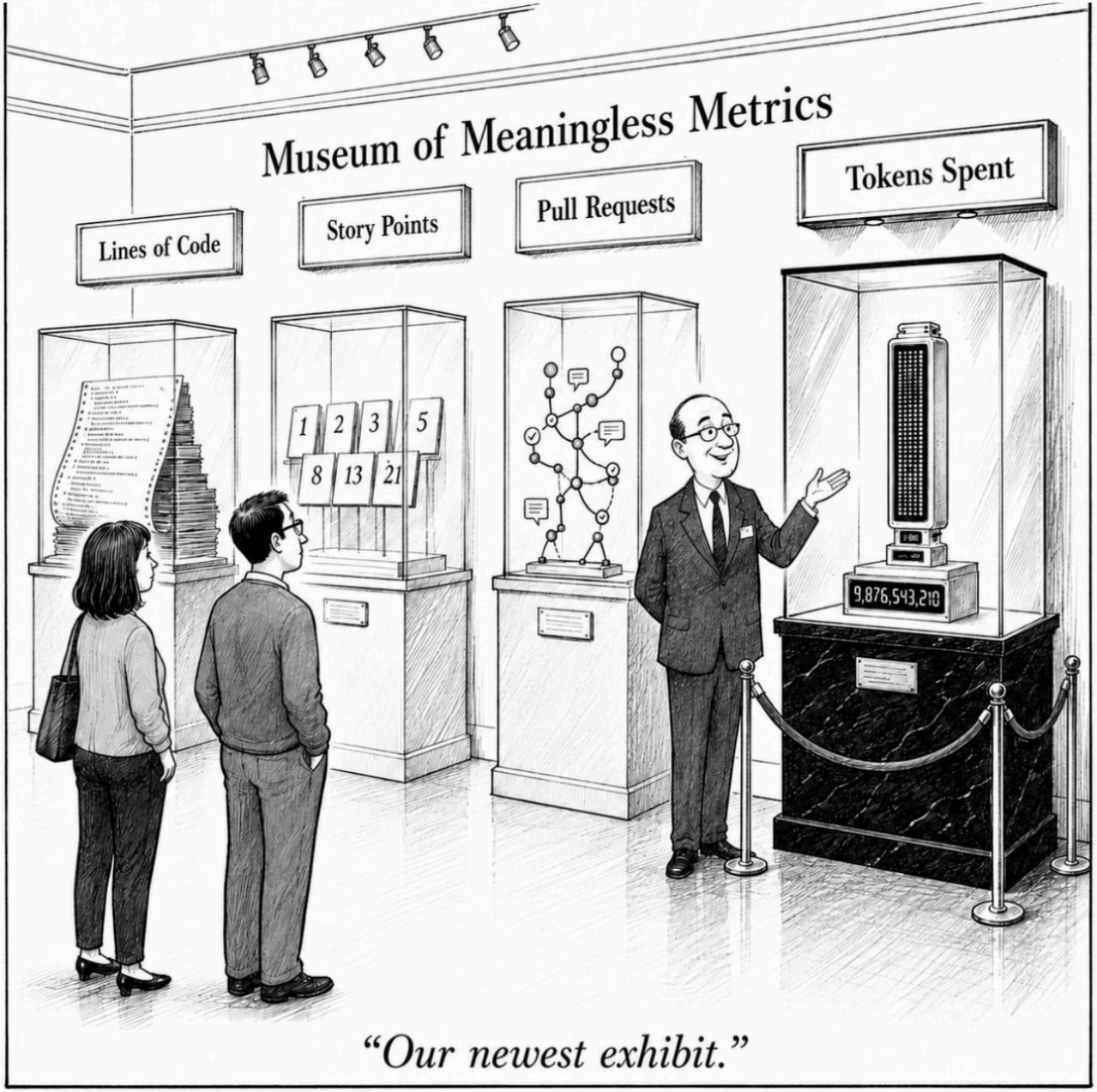
"SonarQube says NOPE !"

"I got to do something to get through this Quality Gate !"

Bad move — Laurie has walked straight into…

Goodhart's Law

> When a measure becomes a target, it stops being a good measure.


Museum of Meaningless Metrics
Lines of Code
Story Points
Pull Requests
Tokens Spent
1
2
3
5
8
13
21
9,876,543,210
"Our newest exhibit."

How do we stop breaking the thermometer?

🤨 Stop saying “Something went wrong — quick! We need a metric!”

🦾 Use a *small* set of team metrics for things the team *controls*

🫂 Review them with *kindness*. A metric is just a chance to improve

Come, we have foosball ! Pastries ! And bore-out too...

So… how's Laurie doing?

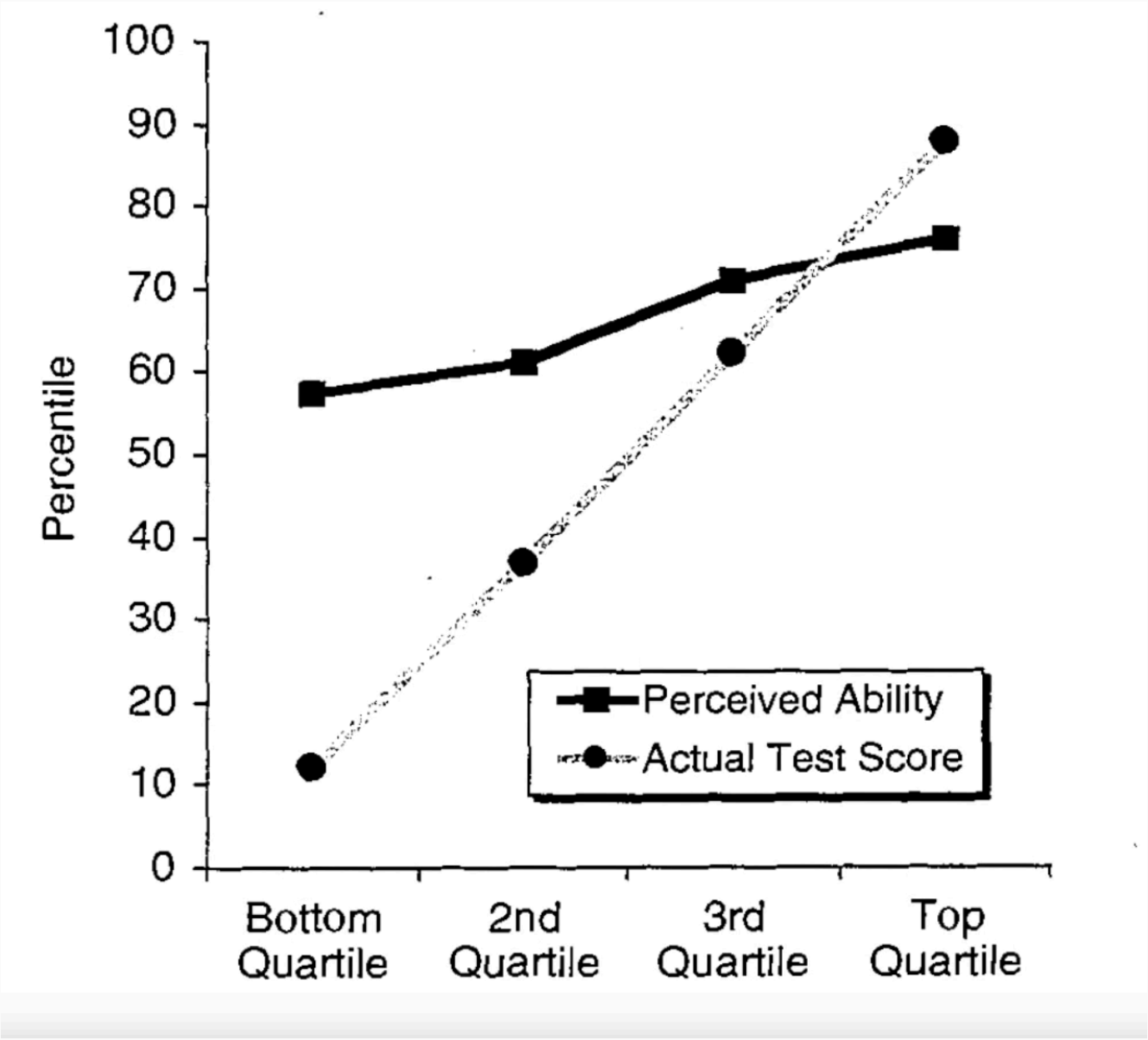
"I have had it with these darned meta-laws in this darned meta-talk !"

The Dunning–Kruger Effect

> People with limited skills in a field often overestimate their abilities.

Corrolary:

> Highly skilled people tend to underestimate their abilities in their own field.


100
90
80
70
60
50
40
30
20
10
0
Percentile
Perceived Ability
Actual Test Score
Bottom Quartile
2nd Quartile
3rd Quartile
Top Quartile

There are three kinds of knowledge:

😌 Things we know

🤔 Things we know we don't know

🤯 Things **we don't know** we don't know

The fix? Easy. Really.

🏃 Keep questioning your own assumptions

👂 Look for skilled people who don't give themselves enough credit

🧘 Watch out for impostor syndrome: trying and failing is learning too

Now Laurie can make a fresh start 😌

“Well, it works on my PowerPoint ! So get to it !”

Come rewrite a monolith in Rust with us !

Hofstadter's Law:

> It always takes longer than you expect, even when you take Hofstadter's Law into account.

Parkinson's Law:

> Work expands to fill the time available for it.

DAVID HEINEMEIER HANSSON

August 19, 2024

Software estimates have never worked and never will

Since the dawn of computing, humans have sought to estimate how long it takes to build software, and for just as long, they've consistently failed. Estimating even medium-sized projects is devilishly difficult, and estimating large projects is virtually impossible. Yet the industry keeps insisting that the method that hasn't worked for sixty years will definitely work on this next project, if we all just try a little harder. It's the definition of delusional.

YO DAWG, I HEARD
YOU WANT FEWER MEETINGS
SO I SCHEDULED MEETINGS TO DISCUSS
THE AMOUNT OF MEETINGS WE SHOULD SCHEDULE
imgflip.com

Éloge de la simplicité - Frédéric Leguédois

Devoxx France
41,9 k abonnés

S'abonner

161

Sooo… now what?

👂 Stop estimating? Tempting! But a tough sell

🏋️ Remember: the “M” in MVP means “Minimum”. Iterate, iterate, iterate

🐷 Give honest (anonymous!) meeting ratings (ROTI). Prefer async communication

“Wait… what did he just say? What's this Lindy Effect?”

"BORING TECHNOLOGIES",
FOR LACK OF A BETTER WORD
ARE GOOD

The Lindy Effect

> For imperishable things (like code), the longer they have existed or been in use, the longer we can expect them to last.

In other words…

🗻 Things that last have a reason to last — and that reason keeps them going

👶 New things haven't faced the real world yet

🙏 New ideas must prove useful… even if they grow old along the way!

💡 “Modern” means “of our time” — not necessarily “better”

With those wise words in mind, Laurie takes on her Cart page again…

Who's a good little complexity ? You are! Yes you are!

Tesler's Law

> An application's inherent complexity can never be fully removed or hidden. You can only move it around and make trade-offs.

In other words…

😅 Your application will always have some complexity

💵 It's in your code (DX), the user experience (UX), the costs (FinOps), etc.

⚖️ Find the right balance and focus on your strengths

DEVOXX MOROCCO

Anyway… Laurie gets back to her application's complexity…

And then…

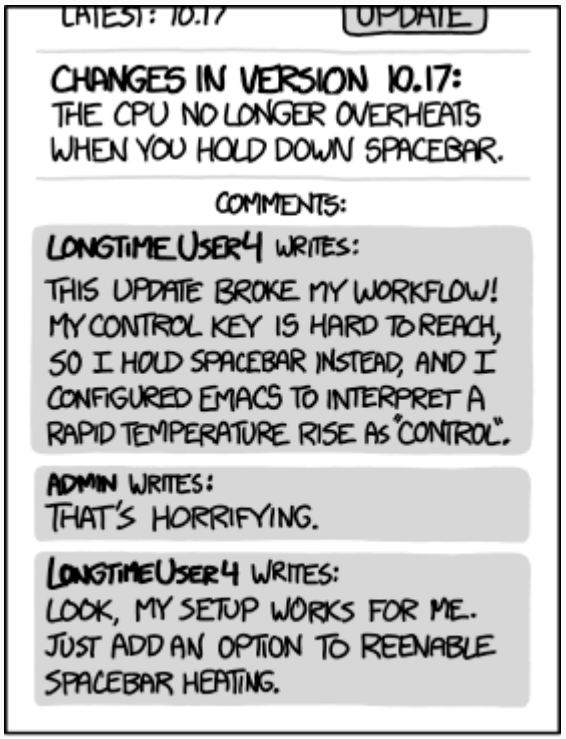
💥 Production bug! The order total in the header says `undefined` 😱

Your worst nightmares? That's his specialty. Of course I'm talking about…

Hyrum's Law — the Accidental Contract

> With enough users of your API, it doesn't matter what your contract promises.

> Someone will depend on every observable behavior of your system.



EVERY CHANGE BREAKS SOMEONE'S WORKFLOW.

Sounds easy enough to handle… on paper:

Expose only what you need to

Tell everyone about changes

SemVer is a **lie**

Test, test, test

DEVOXX MOROCCO

From   Linus Torvalds <>
Date   Sun, 23 Dec 2012 09:36:15 -0800
Subject Re: [Regression w/ patch] Media commit causes user space to misbahave
On Sun, Dec 23, 2012 at 6:08 AM, Mauro Carvalho Chehab
<mchehab@redhat.com> wrote:
>
> Are you saying that pulseaudio is entering on some weird loop if the
> returned value is not -EINVAL? That seems a bug at pulseaudio.

Mauro, SHUT THE FUCK UP!

It's a bug alright - in the kernel. How long have you been a
maintainer? And you *still* haven't learnt the first rule of kernel
maintenance?

If a change results in user programs breaking, it's a bug in the
kernel. We never EVER blame the user programs. How hard can this be to
understand?

To make matters worse, commit f0ed2ce840b3 is clearly total and utter
CRAP even if it didn't break applications. ENOENT is not a valid error
return from an ioctl. Never has been, never will be. ENOENT means "No
such file and directory", and is for path operations. ioctl's are done
on files that have already been opened, there's no way in hell that
ENOENT would ever be valid.

> So, on a first glance, this doesn't sound like a regression,
> but, instead, it looks tha pulseaudio/tumbleweed has some serious
> bugs and/or regressions.

Shut up, Mauro. And I don't _ever_ want to hear that kind of obvious
garbage and idiocy from a kernel maintainer again. Seriously.

I'd wait for Rafael's patch to go through you, but I have another
error report in my mailbox of all KDE media applications being broken
by v3.8-rc1, and I bet it's the same kernel bug. And you've shown
yourself to not be competent in this issue, so I'll apply it directly
and immediately myself.

WE DO NOT BREAK USERSPACE!

Seriously. How hard is this rule to understand? We particularly don't

“Hello Guild! I've opened a PR to update our in-house framework”

AND THEN I TOLD HER
"I'LL LOOK AT YOUR PR BEFORE MY TIME OFF !"

Now here's the perfect example of…

The Shirky Principle

> Institutions will unconsciously try to preserve the problem they are meant to solve.

In other words…

🛑 “Guilds”, “Design Authorities” and “Support” teams that lock everything down

🔒 Setups where DevOps simply can't work (thanks Conway!)

🙄 “It's not us! Support Team XYZ is blocking us!”

DEVOXX MOROCCO

Let's look on the bright side:

😌 Start with the “User”

👮 “Make invalid states impossible to represent”

🎭 Stop the “Security Theater”

My favorite support team 💗

There's still so much more to say about Meta-Laws…

Oh, no, I won't go to your 4h, so-called "Agile 6 month plan" meeting !

Maslow's Hammer

> If the only tool you have is a hammer, everything starts to look like a nail.

Pure bliss, at a cost, though 😅

Stéphane TRÉBEL

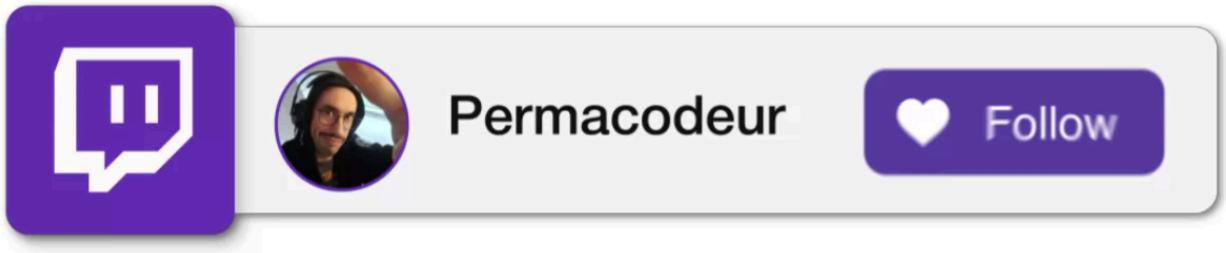
Le Permacodeur

ERP 👔

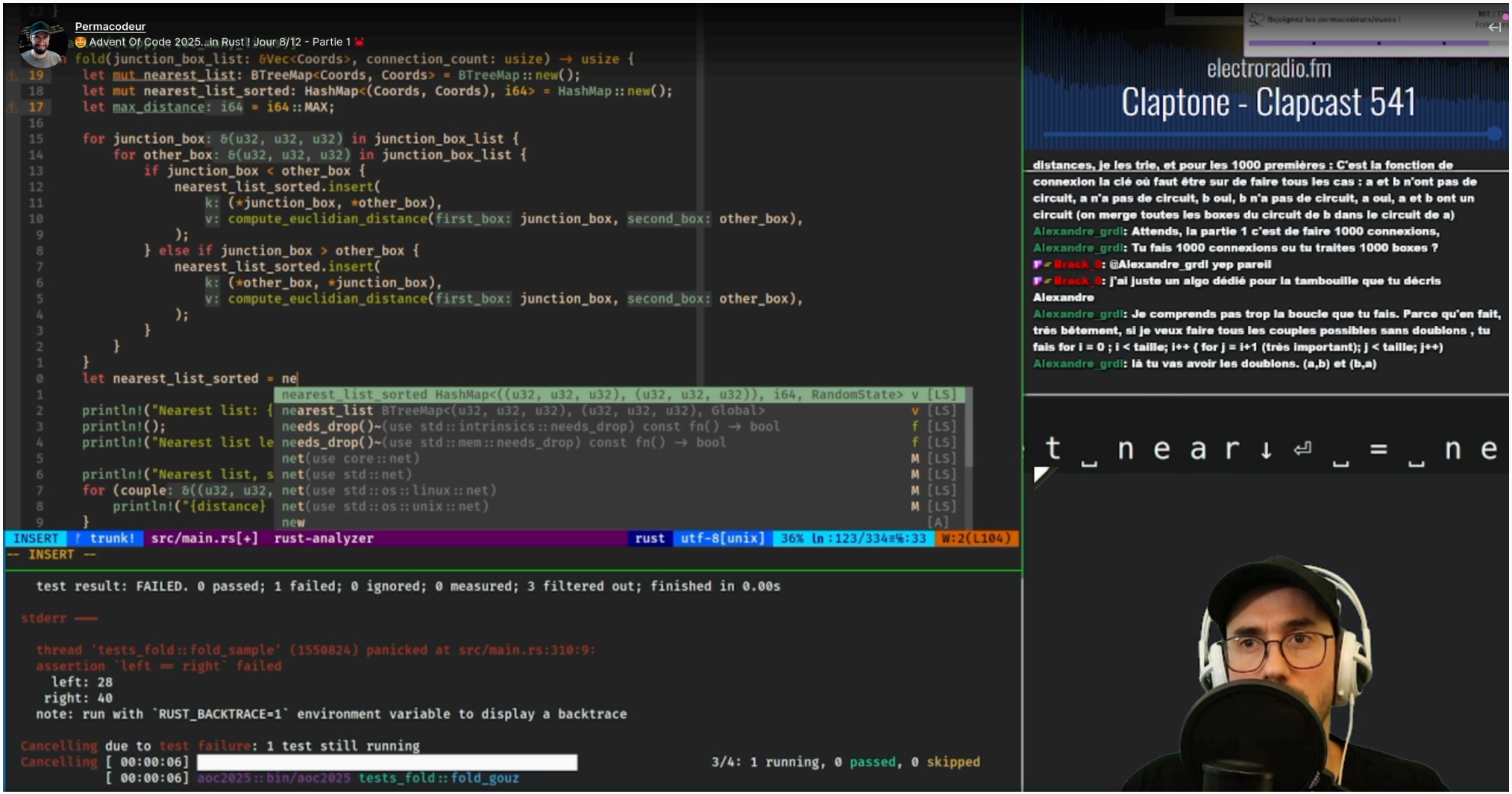
Web 🌍

Rust 🦀

DEVOXX MOROCCO



Like and Subscribe ! 🫶



One of my stream topics: Advent of Code...in Rust ! 🦀

Cabinet
GOURDEL & TRÉBEL Software
BUREAU D'ÉTUDES & D'INGÉNIERIE LOGICIELLE

My newest company, feel free to ask me about it

My links ✍️

Meta-Laws are real. They're out to get you. But now you will know how to fight back ! 😉